

HDOC - Day 34

Matrix Distributive Properties Using Variable-Length 2D Arrays

Algorithm

1. Display the menu. 2. For left distributive property, read m,n,p and matrices A(m x n), B(n x p), C(n x p). Compute B+C, then LHS=A(B+C). Separately compute AB and AC and add them for RHS. 3. For right distributive property, read A(m x n), B(m x n), C(n x p). Compute A+B, then LHS=(A+B)C. Separately compute AC and BC and add them for RHS. 4. Display both m x p result matrices. 5. Compare corresponding elements and state whether the selected property is verified. 6. Repeat until Exit.

Test Case Table

TC	Operation / Dimensions	Expected	Actual	Result
1	Left; m=n=p=2	LHS = RHS	LHS = RHS	Pass
2	Right; m=n=p=2	LHS = RHS	LHS = RHS	Pass
3	Invalid menu choice	Error message	Error message	Pass

C Program

```
#include <stdio.h>

void readMatrix(int r,int c,int a[r][c]);
void addMatrix(int r,int c,int a[r][c],int b[r][c],int x[r][c]);
void multiply(int r1,int c1,int c2,int a[r1][c1],int b[c1][c2],int x[r1][c2]);
void display(int r,int c,int a[r][c]);
int equalMatrix(int r,int c,int a[r][c],int b[r][c]);
void leftDistributive(void);
void rightDistributive(void);

int main() {
    int ch;
    do {
        printf("\n1. Left distributive property\n2. Right distributive property\n3. Exit\n");
        printf("Enter choice: "); scanf("%d",&ch);
        switch(ch) {
            case 1: leftDistributive(); break;
            case 2: rightDistributive(); break;
            case 3: break;
            default: printf("Invalid choice.\n");
        }
    } while(ch!=3);
    return 0;
}

void readMatrix(int r,int c,int a[r][c]) {
    int i,j;
    for(i=0;i<r;i++) for(j=0;j<c;j++) scanf("%d",&a[i][j]);
}

void addMatrix(int r,int c,int a[r][c],int b[r][c],int x[r][c]) {
    int i,j;
    for(i=0;i<r;i++) for(j=0;j<c;j++) x[i][j]=a[i][j]+b[i][j];
}

void multiply(int r1,int c1,int c2,int a[r1][c1],int b[c1][c2],int x[r1][c2]) {
    int i,j,k;
    for(i=0;i<r1;i++)
        for(j=0;j<c2;j++) {
            x[i][j]=0;
            for(k=0;k<c1;k++) x[i][j]+=a[i][k]*b[k][j];
        }
}

void display(int r,int c,int a[r][c]) {
    int i,j;
    for(i=0;i<r;i++) {
        for(j=0;j<c;j++) printf("%d ",a[i][j]);
        printf("\n");
    }
}

int equalMatrix(int r,int c,int a[r][c],int b[r][c]) {
    int i,j;
    for(i=0;i<r;i++) for(j=0;j<c;j++)
        if(a[i][j]!=b[i][j]) return 0;
    return 1;
}

void leftDistributive(void) {
    int m,n,p;
    printf("Enter m n p: "); scanf("%d%d%d",&m,&n,&p);
    int A[m][n],B[n][p],C[n][p],BC[n][p];
    int lhs[m][p],AB[m][p],AC[m][p],rhs[m][p];

    printf("A: "); readMatrix(m,n,A);
    printf("B: "); readMatrix(n,p,B);
    printf("C: "); readMatrix(n,p,C);

    addMatrix(n,p,B,C,BC);
    multiply(m,n,p,A,BC,lhs);
    multiply(m,n,p,A,B,AB);
    multiply(m,n,p,A,C,AC);
    addMatrix(m,p,AB,AC,rhs);

    printf("LHS:\n"); display(m,p,lhs);
    printf("RHS:\n"); display(m,p,rhs);
    if(equalMatrix(m,p,lhs,rhs))
        printf("Left distributive property is verified.\n");
    else printf("Left distributive property is not verified.\n");
}

void rightDistributive(void) {
    int m,n,p;
    printf("Enter m n p: "); scanf("%d%d%d",&m,&n,&p);
    int A[m][n],B[m][n],C[n][p],AB[m][n];
```

```

int lhs[m][p],AC[m][p],BC[m][p],rhs[m][p];

printf("A: "); readMatrix(m,n,A);
printf("B: "); readMatrix(m,n,B);
printf("C: "); readMatrix(n,p,C);

addMatrix(m,n,A,B,AB);
multiply(m,n,p,AB,C,lhs);
multiply(m,n,p,A,C,AC);
multiply(m,n,p,B,C,BC);
addMatrix(m,p,AC,BC,rhs);

printf("LHS:\n"); display(m,p,lhs);
printf("RHS:\n"); display(m,p,rhs);
if(equalMatrix(m,p,lhs,rhs))
    printf("Right distributive property is verified.\n");
else printf("Right distributive property is not verified.\n");
}

```

Compilation, Execution & Testing

```

puppie@Anvikas-MacBook-Air Desktop % clang day34.c -o day34
puppie@Anvikas-MacBook-Air Desktop % ./day34
1. Left distributive property
2. Right distributive property
3. Exit
Enter choice: 1
Enter m n p: 2 2 2
A: 1 2 3 4
B: 5 6 7 8
C: 1 1 1 1
LHS:
10 13
22 29
RHS:
10 13
22 29
Left distributive property is verified.

```

```

puppie@Anvikas-MacBook-Air Desktop % ./day34
Enter choice: 2
Enter m n p: 2 2 2
A: 1 2 3 4
B: 5 6 7 8
C: 1 0 0 1
LHS:
6 8
10 12
RHS:
6 8
10 12
Right distributive property is verified.

```

Result: The program successfully verified the left and right distributive properties of matrix multiplication and displayed matching LHS and RHS matrices.